F79SP Session 12: Stationary distributions

Consider a THMC $\{X_n\}_{n\geq 0}$. Recall a row vector $\boldsymbol{\mu}$ is called a distribution if $\sum_{i\in S}\mu_i = 1$. Given an initial distribution $\boldsymbol{\mu}$ of X_0, we are interested with the distribution of X_n when $n \to \infty$. To study this, we look at two type of distributions, stationary and limiting.

1 Stationary distribution

We consider a motivational example as below.

Example 1.1. *Consider a THMC with state space $S = \{1, 2\}$ with $P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ 1 & 0 \end{pmatrix}$. We are interested with a distribution which stays invariant under the MC's transitions (called stationary distribution) Note that:*

The definition of stationary distributions is given formally as follows.

Definition 1.1. *A (probability) distribution $\boldsymbol{\pi}$ on $\mathcal{S}$ is **stationary** if and only if*

$$\boldsymbol{\pi}P = \boldsymbol{\pi} \qquad \textit{or equivalently} \qquad \sum_{i\in\mathcal{S}} \pi_i p_{ij} = \pi_j, \quad j \in \mathcal{S}. \tag{1}$$

Note that: $\boldsymbol{\pi}P^n = (\boldsymbol{\pi}P)P^{n-1} = \boldsymbol{\pi}P^{n-1} = \cdots = \boldsymbol{\pi}$. We then have also, for all $n \geq 1$,

$$\boldsymbol{\pi}P^n = \boldsymbol{\pi} \qquad \text{or equivalently} \qquad \sum_{i\in\mathcal{S}} \pi_i p_{ij}^{(n)} = \pi_j, \quad j \in \mathcal{S}. \tag{2}$$

Note: If the distribution of X_0 is $\boldsymbol{\pi}$, then, for all $n \geq 1$, the distribution of X_n is also $\boldsymbol{\pi}$.

Example 1.2. *We consider the Bonus-malus system with state space* $S = \{0, 1, 2\}$ *and* $p = 0.1$*, giving:*

$$P = \begin{pmatrix} 0.1 & 0.9 & 0 \\ 0.1 & 0 & 0.9 \\ 0.1 & 0 & 0.9 \end{pmatrix}$$

Determine its stationary distribution.

Result: Stationary distribution need not exist.

Example 1.3. *A SRW has state space $S = \mathbb{Z}$ being a single closed class. We justify that stationary distribution does not exist.*

- *If it were, it must be that $\boldsymbol{\pi} = (c, c, c, \dots)$ for some $0 < c < 1$ due to space homogeneous property.*
- *Note that such $\boldsymbol{\pi}$ cannot be a valid distribution as $\sum_{j \in S} c = \infty \neq 1$.*

Note: Stationary distribution can be viewed as an equilibrium distribution, as once X_n attains a stationary distribution $\boldsymbol{\pi}$, in future, the distribution of the MC will always be $\boldsymbol{\pi}$.